



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : MF F418
Course Title : Lean Manufacturing
Instructor-in-Charge : Abhijeet K. Digalwar

1. Course Description:

Fundamentals of continuous improvement, value-added and waste elimination, elements of lean production: small lot production, setup time reduction, maintaining and improving equipment, pull production systems, focused factories and group technologies, work cells and cellular manufacturing, standard operations, quality of design, systems for eliminating defects, simplified production planning and control systems: scheduling for smooth flow, synchronizing and balancing process, planning and control in pull production, beyond the production systems: managing the supply chain, activity based costing, performance measurement.

2. Scope and Objective of the Course:

In factories around the globe, Toyota consistently raises the bar for manufacturing, product development and process excellence. The result is an amazing business success story: steadily taking market share from price-cutting competitors, earning far more profit than any other automaker, and winning the praise of business leaders worldwide. With a market capitalization greater than the value of General Motors, Ford and Chrysler combined; Toyota is the world's most profitable automaker. Toyota's well known "secret weapon" is lean manufacturing – the revolutionary approach to business processes that it invented in the 1950's and has spent decades for perfecting. Today business around the world is attempting to implement Toyota system to maintain competitive advantage. But are they?

The course explains Toyota's unique approach to Lean manufacturing, that drive Toyota's quality and efficiency obsessed culture. You will valuable insights that can be applied to any organization and any business process, whether in services or manufacturing. You will discover how the right combination of long-term philosophy, processes, people and problem solving can transform your organization into a lean, learning enterprise – the Toyota way.

At the end of this course, the students will be able to understand the importance of lean manufacturing / The Toyota Way as a strategic weapon, the students will be able to understand the World –Class power of the Toyota way

3. Text Books:

TB1. Pascal Dennis, "Lean Production Simplified", 2nd Edition, Productivity Press, 2007.
TB2. Jeffrey K. Liker, "The Toyota Way", McGraw-Hill Edition, New Delhi, 2004.

4. Reference Books:

RB1. James P. Womack and Daniel T. Jones, "Lean Thinking: Banish Waste & Create Wealth in Your Corporation, Revised Edition, Simon & Shuster, 2001.



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RB2. Nicholas, John M., “Competitive Manufacturing Management”, Tata McGraw Hill Education Pvt. Ltd. New Delhi, 2012.

5. Course Plan:

Sr. No	Learning Objectives	Topics to be covered	Text/Reference book	No. of lecture hrs
1	Birth of lean production	Types of production systems, growing dysfunction, birth of lean production, virtue of necessity, lean revolution at Toyota	TB1 Ch1, RB2 Ch1	1-2
2	Lean production system	Why lean production? Systems and systems thinking, basic image of lean production, customer focus, Muda, Mura and Muri, Toyota's Seven Wastes	TB1 Ch2 RB2 Ch3	3-5
3	Stability and Standardized work	Standards in lean system, Continuous Improvement, 5S system, total productive maintenance	TB1 Ch 3 and 4 RB2 Ch 2 and 7	6-14
4	Just-In-Time and Jidoka	Why JIT, principles of JIT, JIT system, pull systems, kanban, kanban rules, Small lot production, setup time reduction, SMED methodology, development of Jidoka concept, why jidoka, expanded role of conveyance, production levelling, Poka Yoke,	TB1 Ch 5 RB2 Ch 5, 6 and 8	15-25
5	Value Stream Mapping	VSM Process, TAKT time, analysis of data, symbols, benefits, Visual Management, Kaizen, Case Studies	TB1 Ch 5 and Class Notes	25-30
6	Involvement and Customer Focus	Why involvement? The terrible waste of humanity, activities supporting involvement, Leadership development, people development, supplier development, kaizen circle activity, practical kaizen training, suggestion programs, what is planning? Why plan? problems with planning, hoshin planning, hoshin planning system, four	TB1 Ch7, 8 and 9 TB2 Ch 15, 16 and 17	31-36



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		phases of hoshin planning, what is lean culture? How does lean culture feel?		
7	14 principles of Toyota way	The world-class power of the Toyota way, Toyota way in action	TB2	37-42
		TOTAL		42

* The lectures may slightly diverge from the aforesaid plan based on students 'background & interest in the topic, which may include special lectures and discussions that would be planned and notified accordingly.

6. Evaluation Scheme:

EC No	Evaluation Components	Nature of Components	Duration	Weightage%	Date & Time
1.	Mid Semester Test	Closed Book	90 minutes	30	10.10.2026 AN1
2.	Quizzes (Surprise) (n—Best n-2)	Closed Book	15 minutes	10	Lecture session
3.	Case Study/group Assignments/Project	Open Book	-	20	
4.	Comprehensive Examination	Closed Book/Open Book	3 hours	40	16.12.2026 FN

7. Chamber Consultation Hour: Friday 5.00 PM

8. Notices: All notices regarding the course will be communicated through Nalanda/ERP/email.

9. Make-up Policy: is not given as a routine. It depends solely on the genuineness of the circumstances under which a student fails to appear in a scheduled evaluation component. Prior permission should be obtained from the Instructor-in-Charge (I/C) in such circumstances. Non-appearance in-class tests/quizzes may lead to NC. **No Make-up for quizzes/surprise tests.**

Instructor-in-charge
MF F418

Instructor's Contact Details

Name	Role	Chamber No	Phone	Ext.	Email
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